3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 3 = 0 ($\because h=10$)
$\therefore$ The condition that for each column having a leading entry, all positions below the leading entry position must contain 0 is True.
$\therefore$ The matrix $\begin{bmatrix} 1 & -3 & 5 \\ 0 & 0 & 8 \\ 0 & 0 & h-10 \end{bmatrix}$ is in echelon form. □

There exists a row in the matrix ($[0\ 0\ 8]$) of the form ($[0\ 0\ 0 \ldots\ldots 0\ b]$)
By Theorem 2, the system of equations is inconsistent, i.e. no solution set exists.
$\therefore$ There does not exist c_1 and c_2 s.t. $c_1v_1+c_2v_2=v_3$. $\therefore v_3 \notin \text{Span}\{v_1,v_2\}$.
$\therefore$ For no h, v_3 belongs to $\text{Span}\{v_1,v_2\}$ (a) □
Suppose $\{v_1,v_2,v_3\}$ are linearly dependent.
$\therefore$ By defn, $\exists c_1,c_2,c_3 \in \mathbb{R}$ s.t. $c_1v_1+c_2v_2+c_3v_3=0$, where not all of c_1,c_2,c_3 are 0 together. By defn. of matrix equation $Ax=b$, we have: $[v_1\ v_2\ v_3]\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}=[0]$
$\begin{bmatrix} 1 & -3 & 5 \\ -3 & 9 & -7 \\ 2 & -6 & h \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. By Theorem 3, the matrix eqn. has the same solution set as the system of equations corresponding to the augmented matrix $\begin{bmatrix} 1 & -3 & 5 & 0 \\ -3 & 9 & -7 & 0 \\ 2 & -6 & h & 0 \end{bmatrix}$

From previous part, the given matrix can be transformed to $\begin{bmatrix} 1 & -3 & 5 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & h-10 & 0 \end{bmatrix}$. The corresponding system of equations are:

① $c_1-3c_2+5c_3=0$ ② $8c_3=0$ ③ $(h-10)c_3=0$
From ②, $c_3=0$ $\therefore c_1-3c_2+5\cdot 0=0 \Rightarrow c_1-3c_2=0 \Rightarrow c_1=3c_2$
Column 2 has no pivots, $\therefore$ by defn, c_2 is a free variable. If $c_2=1$, $c_1=3$.
$\therefore 3\begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}+1\cdot\begin{bmatrix} -3 \\ 9 \\ -6 \end{bmatrix}+0\cdot\begin{bmatrix} 5 \\ -7 \\ h \end{bmatrix}=0$, holds for all h □

In Exercises 11–14, find the values of h for which the vectors are linearly dependent. Justify each answer.